

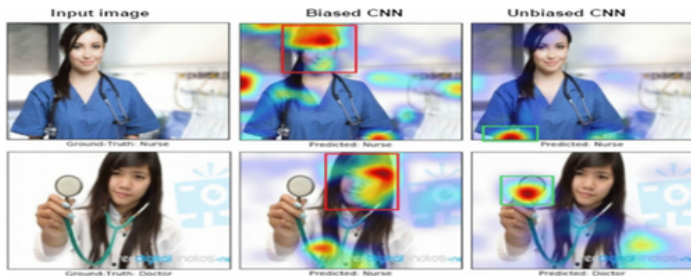
Quantifying Explainability and Applying it to Quanvolutional Neural Networks

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Problem Statement

What numbers would make you trust your neural network?



Visualizations (Selvaraju et al., 2017) do not provide a comprehensive overview of the explainability of a model.

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- Q1 Is there a way to quantify the explainability of neural networks that provides average and class-wise explainabilities?
- Q2 How to demonstrate the effectiveness of the proposed method?

Quantifying Explainability

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Paper	Citations	Models	Description
An objective metric for explainable AI (Sovrano and Vitali, 2023)	39	Regression & ANN for healthcare & finance	NLP-Based: Evaluate answers extracted by pre-trained LM for specific questions
Quantifying explainers of graph neural networks in computational pathology (Jaume et al., 2021)	107	Graph NN for Computational Pathology	Statistics of class separability using clinical features to characterize graph explainers
Explainable deep classification models for domain generalization (Zunino et al., 2021)	52	Deep CNN for object detection	Guided model training by incorporating feedback from ideal heatmaps to enhance the explainability of DNNs.
Towards Quantification of Explainability in Explainable Artificial Intelligence Methods (Islam, Eberle, and Ghafoor, 2020)	71	Supervised ML Models for mortgage bankruptcy prediction	Analyzed the definitions, quantified explainability in terms of faithfulness, stability, and comprehensibility of an explanation.

Quanvolutional Neural Networks

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Quanvolutional neural networks (Henderson et al., 2020)	418	MNIST	Random quantum circuits used to encode images in QNNs achieved performance comparable to CNNs
Variational quanvolutional neural networks with enhanced image encoding (Mattern et al., 2021)	27	MNIST	The effectiveness of image encodings in trainable VQNNs varies; the choice depends on a specific application.
RANet: a custom CNN and QNN for the automated detection of rheumatoid arthritis Ahalya et al., 2023	15	Real-time hand thermograms	CNN and QNN achieved similar accuracy, showing they could be employed as a pre-screening diagnostic tool.
Advqunn: A methodology for analyzing the adversarial robustness of quanvolutional neural networks (El Maouaki et al., 2024)	9	MNIST & FM-NIST	QNNs found to be more robust against adversarial attacks than CNNs

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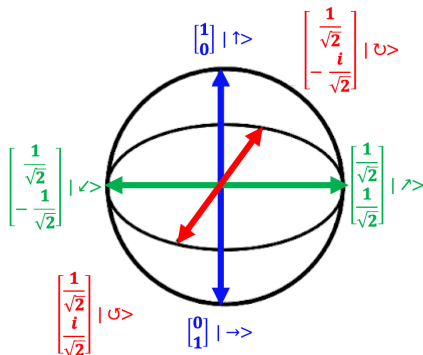
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- Classical bit: 0 or 1
- Qubit:

$$\alpha|0\rangle + \beta|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

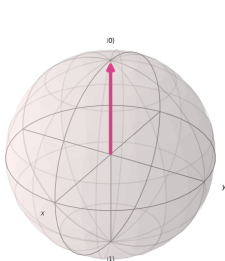


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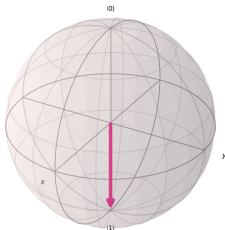
¹“Quantum Computing, Zero to Hero: Part Four — The Qubit & The Bloch Sphere”. In: (). URL: <https://medium.com/@gwilymnewton/quantum-computing-zero-to-hero-part-four-the-qubit-the-bloch-sphere-5200a6e473af>.

Quantum Computing

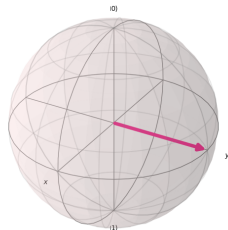
Single-Qubit gates rotate qubits



Qubit originally in
state $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$



Rotating $|0\rangle$ by π
radian around Y-axis
using **RY gate** to
obtain $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$



Rotating $|0\rangle$ by $\frac{\pi}{2}$
radian around X-axis
using **RX gate** to
obtain $|i\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$

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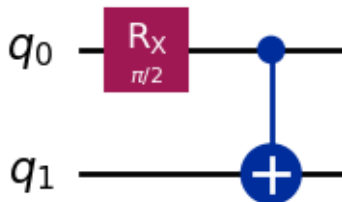
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Two-qubit gates entangle two different qubits

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$



This circuit is used to generate the state described above

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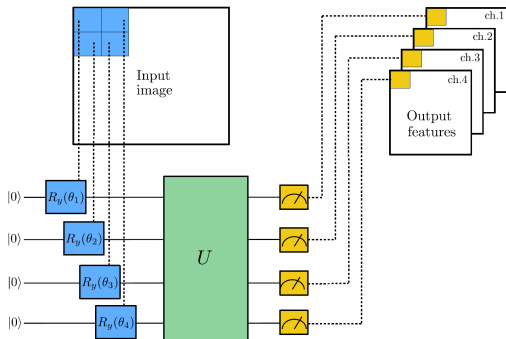
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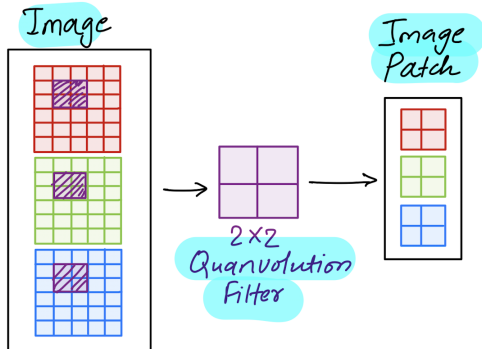
Quantum Convolutional Neural Network



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²"PennyLane tutorial: Quantum Convolutional Neural Networks". In: (). URL: https://pennylane.ai/qml/demos/tutorial_quantumconvolution.

QNN: Input



The Quanvolutional kernel is applied across the input RGB image in a manner similar to how convolutional kernels are used in CNNs.

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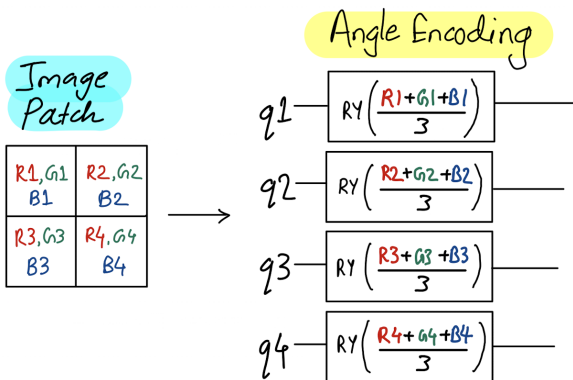
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QNN: Encoding

Two ways of encoding 2x2 image patch into a 4-qubit quantum circuit



Rotate each qubit by averaging RGB pixel values

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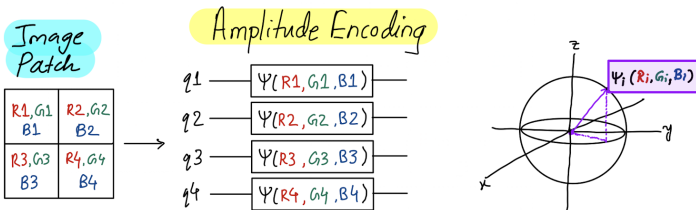
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QNN: Encoding

Two ways of encoding 2x2 image patch into a 4-qubit quantum circuit



Use RGB pixel as (x, y, z) co-ordinates to map them on a on Bloch sphere

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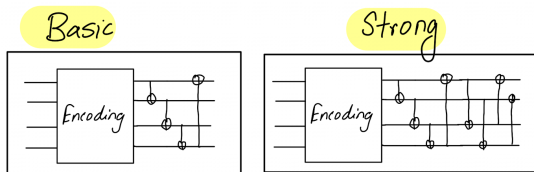
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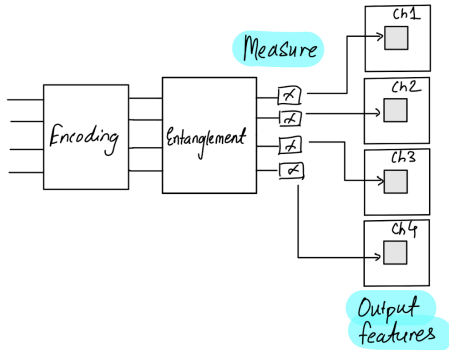
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QNN: Entanglement

- Apply entangling gates (CNOTs) to model dependencies between different pixels of the image.
- Stronger entanglement allows the quantum system to capture deeper and more intricate dependencies that classical models might miss (Ghazi Vakili et al., 2025).



QNN: Measurement



Finally, the qubits are measured to produce different quanvolutional channels.

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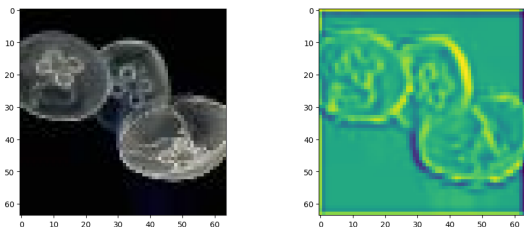
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Heatmap generation

- Methods like Grad-CAM (Gradient-weighted Class Activation Mapping Selvaraju et al., 2017) visualizes a heatmap.
- Heatmap is a 2D matrix showing the importance of each image region, i.e., feature.
- **Higher values (brighter pixels) mean the feature played high importance in the model's decision**



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- The explainability $\mathcal{E}_{NN}$ of a neural network is quantified using the Frobenius norm.
- The Frobenius norm **captures the similarity between the difference of the heatmap of a neural network and an ideal heatmap.**

$$\mathcal{E}_{NN} = ||L_{Ideal} - L_{NN}||_F$$

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$$\mathcal{E}_{NN} = ||L_{Ideal} - L_{NN}||_F$$

- L_{NN} : **heatmap of NN** generated using algorithms like Grad-CAM (Selvaraju et al., 2017).
- L_{Ideal} : **ideal heatmap** showing what features should be used to make classifications
- L_{Ideal} can be generated using existing highly explainable neural networks (Ibrahim and Shafiq, 2023).

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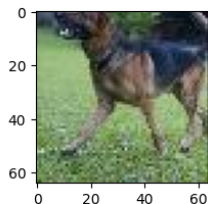
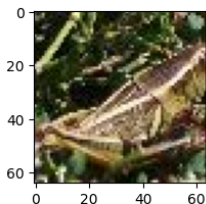
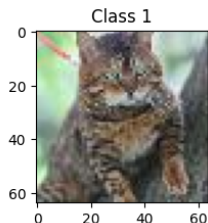
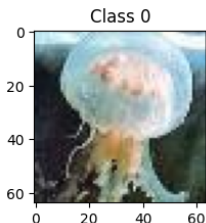
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- **Tiny-ImageNet** selected which fulfils following goals:
 - ① Small and fits with limited computational resources
 - ② Attaining the aim of analyzing explainability for real-world situations
- 200 classes, 500 64x64 RGB images in each class

Dataset

Four classes (**Jellyfish**, **Cat**, **Dog**, **Grasshopper**) chosen for classification and demonstration of explainability



Class 2

Class 3

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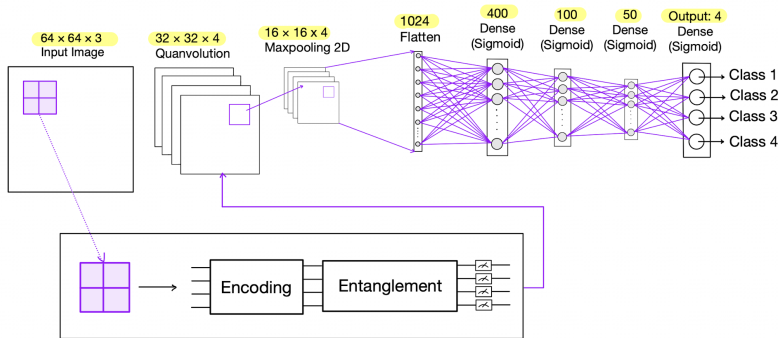
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QNN Architecture



Full architecture of the Quanvolutional Neural Network

QNN Hyperparameters

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- GridSearchCV to find the optimal optimizer from (**adam**, **nadam**, **sgd**) with **two-fold cross-validation**.
- **Sigmoid** found to perform best with manual experiments compared to (**softmax**, **tanh**, **relu**)
- QNN models under different encoding and entanglement schemes trained separately with:
 - Training epochs: 500
 - Batch size: 32
 - Train-test split: 75% random

Ideal Heatmap Generation

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- L_{Ideal} generated using CNNs proposed in Tavanaei, 2020 which have **encoder-decoder to represent regions of interest in an image**.
- Tavanaei, 2020 performed experiments on CIFAR, MNIST, and Tiny-ImageNet with their pre-trained models available on GitHub.

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Evaluate the following **overall** and **class-wise** metrics under different QNN configurations **from 10 different training runs**:

- 1 Accuracy
- 2 F1-Score
- 3 Explainability $\mathcal{E}_{\text{QNN}}$

Results

Overall accuracy, F1-Score and $\mathcal{E}_{QNN}$ of QNNs under different configurations.

Encoding	Entanglement	Accuracy	F1-Score	$\mathcal{E}_{QNN}$
Angle	Basic	0.8793 ± 0.006	0.9355 ± 0.003	22.7294 ± 0.196
Angle	Strong	0.8741 ± 0.008	0.9324 ± 0.004	22.4674 ± 0.199
Amplitude	Basic	0.8757 ± 0.025	0.9334 ± 0.014	24.0631 ± 0.476
Amplitude	Strong	0.8759 ± 0.032	0.9332 ± 0.02	23.5621 ± 0.326

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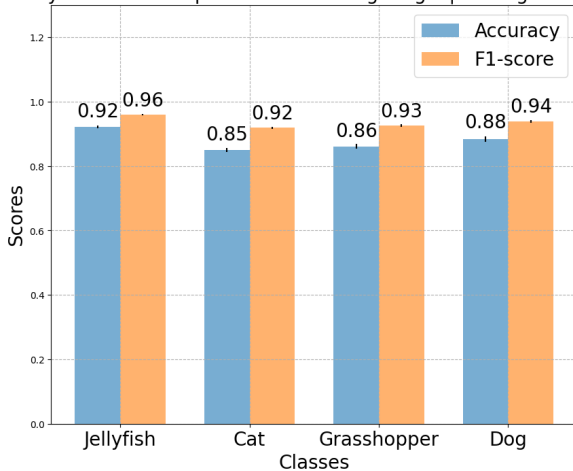
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Class-wise Accuracy & F1-Score comparison

Accuracy and F1-Score per class: Encoding: angle | Entanglement: basic



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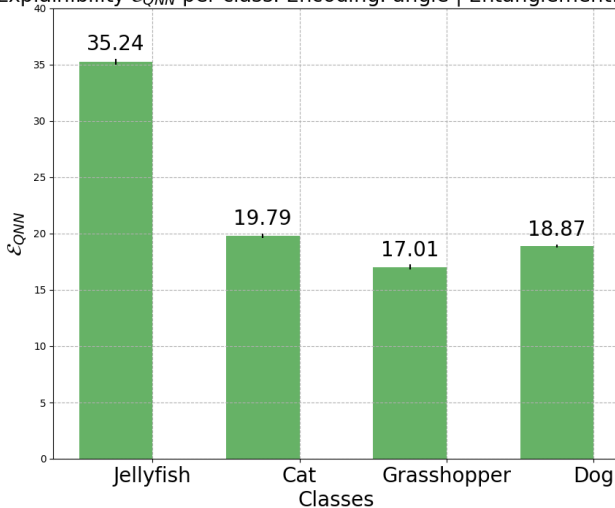
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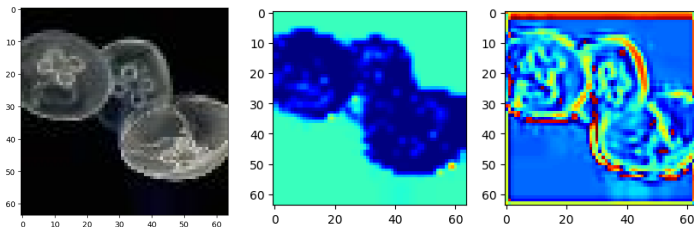
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Class-wise $\mathcal{E}_{QNN}$ comparison

Explainability $\mathcal{E}_{QNN}$ per class: Encoding: angle | Entanglement: basic



Results



Left: Original Jellyfish image. **Middle:** Grad-CAM heatmap for QNN (angle encoding and basic entanglement), which relies on the black background to make the classification. **Right:** The ideal heatmap is generated using encoder-decoder-based CNN (Tavanaei, 2020).

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- First-ever analysis of QNN's explainability.
- High accuracy and high F1-score do not equate to high explainability.
- The proposed explainability quantification is intuitive and applicable to a wide variety of neural networks.
- E.g., the most crucial application in healthcare to analyze if a model is focusing on the right features to make decisions. Collaboration with medical experts can help generate ideal heatmaps.

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- Experiment with larger quanvolutional neural networks
- Experiment with other complex datasets (CIFAR, ImageNET, etc.)
- Research on how to make QNN more explainable and asses their scalability

GridSearch CV Results

GridSearchCV scores for (NADAM, ADAM, SGD) optimizer under different QNN configurations

Encoding	Entanglement	Optimizer	Best-Score
Angle	Basic	NADAM	0.4476
Angle	Strong	ADAM	0.471
Amplitude	Basic	NADAM	0.542
Amplitude	Strong	NADAM	0.543

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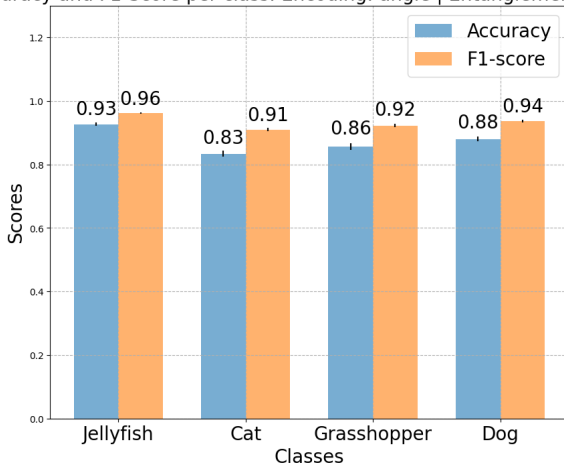
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Angle Encoding & Strong Entanglement

Accuracy and F1-Score per class: Encoding: angle | Entanglement: strong



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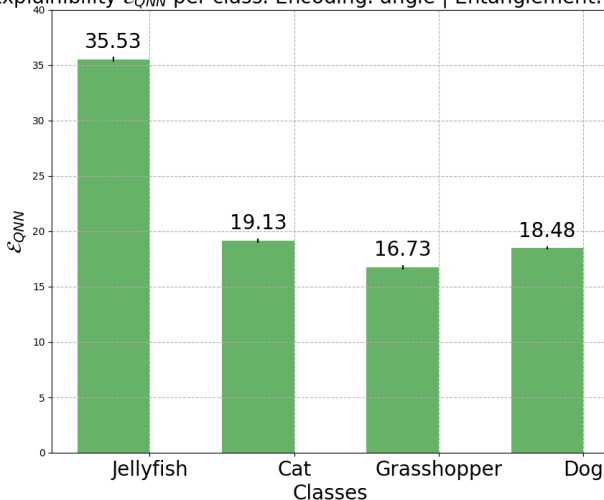
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Angle Encoding & Strong Entanglement

Explainability $\mathcal{E}_{QNN}$ per class: Encoding: angle | Entanglement: strong



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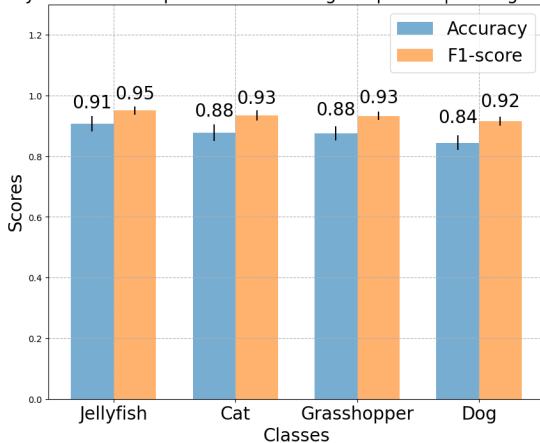
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Amplitude Encoding & Basic Entanglement

Accuracy and F1-Score per class: Encoding: amplitude | Entanglement: basic



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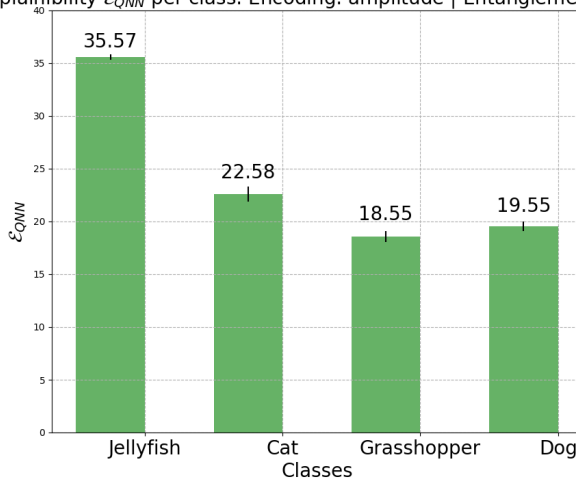
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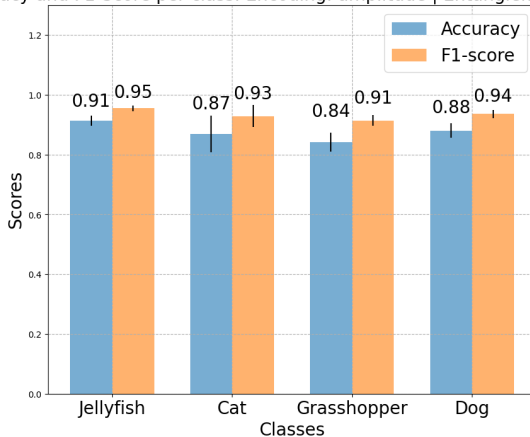
Amplitude Encoding & Basic Entanglement

Explainability $\mathcal{E}_{QNN}$ per class: Encoding: amplitude | Entanglement: basic



Amplitude Encoding & Strong Entanglement

Accuracy and F1-Score per class: Encoding: amplitude | Entanglement: strong



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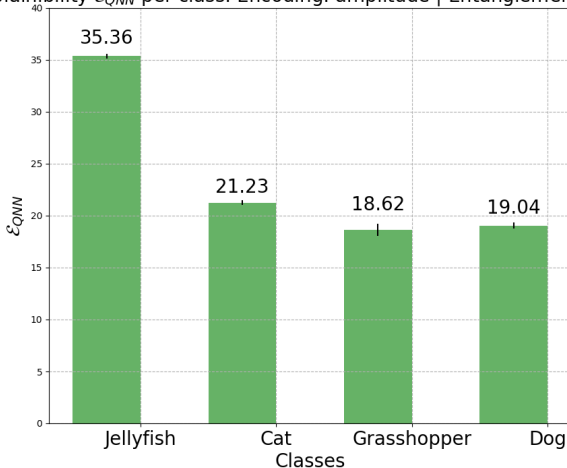
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Explainability $\mathcal{E}_{QNN}$ per class: Encoding: amplitude | Entanglement: strong



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